Q1 a) Write a program to declare class 'student' having data members as roll_no and name and class. A

```
#include <iostream>
#include <string>
using namespace std;
class Student{
public:
    int roll_no;
    string name;
    string classname;
    void accept(){
        cin>>roll_no>>ws;
        getline(cin,name);
        getline(cin,classname);
    }
    void display(){
        cout<<roll_no<<"\n"<<name<<"\n"<<classname<<"\n";
    }
};
int main(){
    Student s;
    s.accept();
    s.display();
    return 0;
}
```

Q1 b) Write a program to declare class 'Book' having data members as book_name, price and no. of pag

```
#include <iostream>
#include <string>
using namespace std;
class Book{
public:
    string name;
    double price;
    int pages;
    void accept(){
        getline(cin,name);
        cin>>price>>pages;
        cin.ignore();
    }
    void display(){
        cout<<name<<"\n"<<price<<"\n"<<pages<<"\n";
    }
};
int main(){
    Book b1,b2;
    b1.accept();
    b2.accept();
    if(b1.price>b2.price) b1.display(); else b2.display();
    return 0;
}
```

Q1 c) Write a program to declare class 'time', accept time in HH:MM:SS format, convert it into total seco

```
#include <iostream>
using namespace std;
class Time{
public:
```

```
    int h,m,s;
    void accept(){ char c; cin>>h>>c>>m>>c>>s; }
    void display(){
        int total = h*3600 + m*60 + s;
        cout<<total<<"\n";
    }
};
int main(){
    Time t;
    t.accept();
    t.display();
    return 0;
}
```

Q2 a) WAP to declare a class 'city' having data members as name and population. Accept this data for 5

```
#include <iostream>
#include <string>
using namespace std;
class City{
public:
    string name;
    long population;
    void accept(){ getline(cin,name); cin>>population; cin.ignore(); }
    void display(){ cout<<name<<" "<<population<<"\n"; }
};
int main(){
    City arr[5];
    for(int i=0;i<5;i++) arr[i].accept();
    for(int i=0;i<5;i++) arr[i].display();
    return 0;
}
```

Q2 b) WAP to declare class 'Account' having data members as Account no and balance. Accept data fo

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;
class Account{
public:
    int accno;
    double balance;
    void accept(){ cin>>accno>>balance; }
    void display(){ cout<<accno<<" "<<balance<<"\n"; }
};
int main(){
    vector<Account> a(10);
    for(int i=0;i<10;i++) a[i].accept();
    for(int i=0;i<10;i++){
        if(a[i].balance>=5000) a[i].balance += a[i].balance*0.10;
    }
    for(int i=0;i<10;i++) a[i].display();
    return 0;
}
```

Q2 c) WAP to declare class 'Staff' having data members as name and post. Accept this data for 5 staf

```
#include <iostream>
#include <string>
using namespace std;
class Staff{
public:
    string name;
    string post;
    void accept(){ getline(cin,name); getline(cin,post); }
};
int main(){
    Staff s[5];
    for(int i=0;i<5;i++) s[i].accept();
    for(int i=0;i<5;i++) if(s[i].post=="HOD") cout<<s[i].name<<"\n";
    return 0;
}
```

Q3 a) Write a program to declare a class 'book' containing data members as book-title, author, name an

```
#include <iostream>
#include <string>
using namespace std;
class Book{
public:
    string title;
    string author;
    double price;
    void accept(){ getline(cin,title); getline(cin,author); cin>>price; cin.ignore(); }
    void display(){ cout<<title<<"\n"<<author<<"\n"<<price<<"\n"; }
};
int main(){
    Book b;
    Book *p=&b;
    p->accept();
    p->display();
    return 0;
}
```

Q3 b) WAP to declare class 'student' having data members roll_no and percentage. Using 'this' pointer

```
#include <iostream>
#include <string>
using namespace std;
class Student{
public:
    int roll_no;
    float percentage;
    void accept(){ cin>>roll_no>>percentage; }
    void display(){ cout<<roll_no<<" "<<percentage<<"\n"; }
    void invoke(){ this->display(); }
};
int main(){
    Student s;
    s.accept();
    s.invoke();
    return 0;
}
```

Q4 a) WAP to create two class results and result2 which stores the marks of the students. Read value

```
#include <iostream>
using namespace std;
class Result{
public:
    int marks1;
    int marks2;
    void accept(){ cin>>marks1>>marks2; }
    int total(){ return marks1+marks2; }
};
int average(Result r1, Result r2){
    return (r1.total()+r2.total())/2;
}
int main(){
    Result a,b;
    a.accept();
    b.accept();
    cout<<average(a,b)<<"\n";
    return 0;
}
```

Q4 b) WAP to find the greatest number among two numbers from two different classes using friend fun

```
#include <iostream>
using namespace std;
class A{
public:
    int x;
    void accept(){ cin>>x; }
    friend int greatest(A,A);
};
int greatest(A a1,A a2){
    return (a1.x>a2.x)?a1.x:a2.x;
}
int main(){
    A p,q;
    p.accept();
    q.accept();
    cout<<greatest(p,q)<<"\n";
    return 0;
}
```

Q4 c) WAP for swapping contents of two variables of same class using friend function.

```
#include <iostream>
using namespace std;
class Num{
public:
    int v;
    void accept(){ cin>>v; }
    friend void swapNum(Num&,Num&);
};
void swapNum(Num &a, Num &b){
    int t=a.v; a.v=b.v; b.v=t;
}
int main(){
    Num a,b;
    a.accept();
```

```
    b.accept();
    swapNum(a,b);
    cout<<a.v<<" "<<b.v<<"\n";
    return 0;
}
```

Q5 a) Write a program to find the sum of numbers between 1 to n using a constructor where the value o

```
#include <iostream>
using namespace std;
class SumN{
public:
    int n;
    int s;
    SumN(int x){ n=x; s=0; for(int i=1;i<=n;i++) s+=i; }
    void display(){ cout<<s<<"\n"; }
};
int main(){
    int n; cin>>n;
    SumN obj(n);
    obj.display();
    return 0;
}
```

Q5 b) Write a program to declare a class 'student' having data members as name and percentage. Write

```
#include <iostream>
#include <string>
using namespace std;
class Student{
public:
    string name;
    int percentage;
    Student(string n,int p){ name=n; percentage=p; }
    void display(){ cout<<name<<"\n"<<percentage<<"\n"; }
};
int main(){
    string name; int p;
    getline(cin,name);
    cin>>p;
    Student s(name,p);
    s.display();
    return 0;
}
```

Q5 c) Define a class 'College' members variables as roll_no, name, course. Use constructor with defaul

```
#include <iostream>
#include <string>
using namespace std;
class College{
public:
    string name;
    string course;
    College(string n,string c){ name=n; course=c; }
    void display(){ cout<<name<<" "<<course<<"\n"; }
};
int main(){
    string n1,c1,n2,c2;
```

```
    getline(cin,n1); getline(cin,c1);
    getline(cin,n2); getline(cin,c2);
    College cobj1(n1,c1), cobj2(n2,c2);
    cobj1.display();
    cobj2.display();
    return 0;
}
```

Q6 a) Write a program to implement single inheritance.

```
#include <iostream>
using namespace std;
class Base{ public: int a; void accept(){ cin>>a; } };
class Derived: public Base{ public: int b; void accept2(){ cin>>b; } void display(){ cout<<a<<" "<<b
int main(){ Derived d; d.accept(); d.accept2(); d.display(); return 0; }
```

Q6 b) Write a program to implement multilevel inheritance.

```
#include <iostream>
using namespace std;
class A{ public: int a; void accept(){ cin>>a; } };
class B: public A{ public: int b; void accept2(){ cin>>b; } };
class C: public B{ public: int c; void accept3(){ cin>>c; } void display(){ cout<<a<<" "<<b<<" "<<c<
int main(){ C obj; obj.accept(); obj.accept2(); obj.accept3(); obj.display(); return 0; }
```

Q6 c) Write a program to implement multiple inheritance.

```
#include <iostream>
using namespace std;
class A{ public: int a; void accept(){ cin>>a; } };
class B{ public: int b; void accept2(){ cin>>b; } };
class C: public A, public B{ public: void display(){ cout<<a<<" "<<b<<"\n"; } };
int main(){ C obj; obj.accept(); obj.accept2(); obj.display(); return 0; }
```

Q6 d) Write a program to implement hierarchical inheritance.

```
#include <iostream>
using namespace std;
class Parent{ public: int p; void acceptp(){ cin>>p; } };
class Child1: public Parent{};
class Child2: public Parent{};
int main(){ Child1 c1; Child2 c2; c1.acceptp(); c2.acceptp(); cout<<c1.p<<" "<<c2.p<<"\n"; return 0;
```

Q6 e) Write a program to implement hybrid inheritance.

```
#include <iostream>
using namespace std;
class A{ public: int a; void accept(){ cin>>a; } };
class B: public A{ public: int b; void accept2(){ cin>>b; } };
class C: public A{ public: int c; void accept3(){ cin>>c; } };
int main(){ B b; C c; b.accept(); b.accept2(); c.accept(); c.accept3(); cout<<b.a<<" "<<b.b<<" "<<c.
```